Simple linear regression CHEAT SHEET

THE MODEL AND THE ESTIMATE

model: $y = \beta_0 + \beta_1 x + \varepsilon$ (ε: everything else)

mean of y: $E(y) = \beta_0 + \beta_1 x$ ($\beta_1 = 0$: x tells nothing)

estimate: $\hat{y} = b_0 + b_1 x$ residual $e = y - \hat{y}$

warehouse (orders x, $k cost y): $\hat{y} = 40 + 4.5x$

b_1: +$4.50 an order; b_0: outside the data (x 35–65)

predict inside the data only: $\hat{y}(58) = 40 + 261 = 301$

LEAST SQUARES

least squares: minimise $\Sigma(y - \hat{y})^2$

$$b_1 = \frac{\Sigma(x - \bar{x})(y - \bar{y})}{\Sigma(x - \bar{x})^2} = \frac{4608}{1024} = 4.5$$

$$b_0 = \bar{y} - b_1\bar{x} = 265 - 4.5 \times 50 = 40$$

the line passes through $(\bar{x}, \bar{y})$

$b_1 = r \cdot s_y/s_x = 0.954 \times 50.3/10.7 \approx 4.5$

SUMS OF SQUARES, R² AND R

$SST = \Sigma(y - \bar{y})^2 = 22{,}784$ $SSE = \Sigma(y - \hat{y})^2 = 2{,}048$

$SSR = \Sigma(\hat{y} - \bar{y})^2 = 20{,}736$ $SST = SSR + SSE$

$$r^2 = \frac{SSR}{SST} = \frac{20{,}736}{22{,}784} \approx 0.910 \quad \text{(RSQ)}$$

$r = \pm\sqrt{r^2}$, sign of b_1: +0.954 (CORREL)

high r^2 is not cause, and not proof of a straight line:

$y = x^2$ on 1…8 gives $r^2 \approx 0.953$ (plot the residuals)

ASSUMPTIONS AND TESTS

assume ε: mean 0, one variance σ^2, independent, normal

$$s^2 = MSE = \frac{SSE}{n - 2} = \frac{2048}{8} = 256;\ s = 16 \text{ (STEYX)}$$

$$s_{b_1} = \frac{s}{\sqrt{\Sigma(x - \bar{x})^2}} = \frac{16}{32} = 0.5$$

$$H_0: \beta_1 = 0.\ t = \frac{b_1}{s(b_1)} = 9, \text{ df } 8, p \approx 0.00002$$

β_1: $b_1 \pm t_{\alpha/2}\, s_{b_1} = 4.5 \pm 2.306 \times 0.5 = 3.35 \ldots 5.65$

$F = MSR/MSE = 20{,}736/256 = 81 = t^2$ (one x)

significant ≠ causal ≠ important: read the interval

OUTPUT: LINEST AND ANOVA

b_1 4.5	b_0 40
$se(b_1)$ 0.5	$se(b_0)$ 25.51
r^2 0.9101	s 16
F 81	df 8
SSR 20,736	SSE 2,048

source	SS	df	MS	F
regression	20,736	1	20,736	81
error	2,048	8	256	
total	22,784	9		

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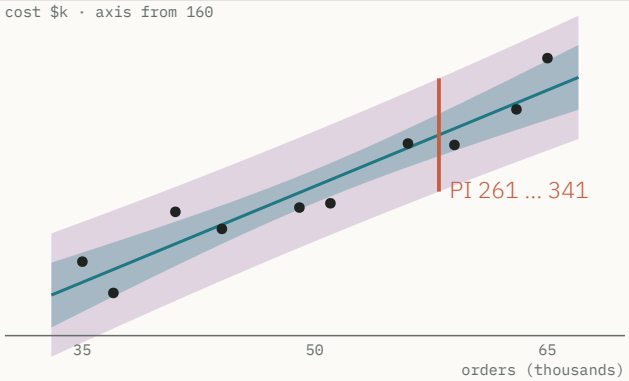
CONFIDENCE AND PREDICTION



at $x^* = 58$: $\hat{y} = 301$ ($k)

mean of y: $s_{\hat{y}} = s\sqrt{(1/n + (x^* - \bar{x})^2/\Sigma(x - \bar{x})^2)} = 6.45$

95 % CI 301 ± 14.9 → 286.1 … 315.9

one new y: $s_{pred} = s\sqrt{(1 + 1/n + \ldots)} = 17.25$

95 % PI 301 ± 39.8 → 261.2 … 340.8

budget for the week, not the average: plan to 341

both narrowest at $\bar{x}$; PI never below ≈ $\pm 2\sigma$

CHECK BEFORE YOU TRUST IT

before you trust a regression:

1 plot the data, then the residuals

2 check leverage (6/n) and standardized e (±2)

3 refit without any flagged point

4 mean of y → CI; one new value → PI

5 stay inside the range of x

RESIDUAL ANALYSIS

plot e against x: band ✓ · funnel × variance

curve × linearity (fuel vs speed: $r^2 \approx 0.88$, yet curved)

in time order: drift × independence

standardized: $\frac{e}{s\sqrt{(1 - h)}}$; look beyond ±2

busiest week: $22.5/(16\sqrt{0.68}) \approx 1.70$

normal plot: sorted standardized e vs normal scores

OUTLIERS, LEVERAGE, INFLUENCE

leverage $h = 1/n + (x - \bar{x})^2/\Sigma(x - \bar{x})^2$; mean 2/n

flag $h > 6/n$ (3 × the mean)

outlier at $\bar{x}$: slope kept, s 16 → 25.6

holiday week $x = 95$: $h \approx 0.675 > 6/11 \approx 0.545$

on the line: b_1 4.51; off it: b_1 5.32

its standardized e only 1.76: influence hides

refit without it; report both; investigate first

CASE: BRANNOCK ENGINEERING (FICTIONAL)

122 staff: 120 engineers, 2 directors (board pay)

average − $20k: 23 flagged, 18 of them juniors

all staff: b_1 4.02, s 14.5, h 0.09 & 0.12: no flags

engineers: $\hat{y} = 69.0 + 3.13x$, r^2 0.81, s 8.7

5 below −2: ≈ $102.5k a year to the line

the CI band would flag 47: wrong yardstick

SPREADSHEET AND PYTHON

SLOPE(y, x) · INTERCEPT(y, x) · RSQ · CORREL

STEYX(y, x) = s · DEVSQ(x) = $\Sigma(x - \bar{x})^2$

FORECAST.LINEAR(x*, y, x) · TREND(y, x, new x)

LINEST(y, x, TRUE, TRUE): the 5 × 2 block

T.INV.2T(0.05, n − 2) · T.DIST.2T · F.DIST.RT

y comes first in every spreadsheet function

Python: scipy.stats.linregress(x, y): slope, intercept, rvalue, pvalue, stderr